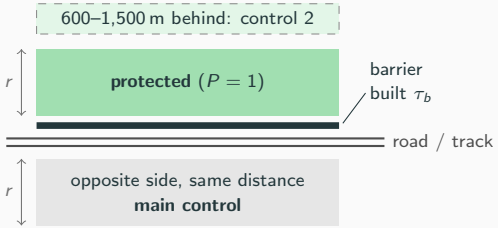
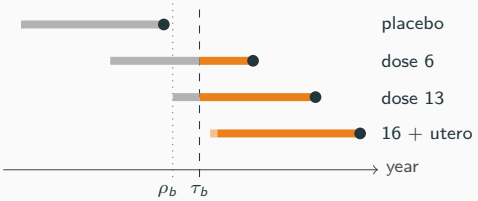
Design: two sides of a road, before and after the barrier

One corridor per barrier



Birth cohorts around τ_b



Bars run from birth to the test at 16 (•); brown marks years with the barrier.

Difference-in-differences within each barrier

- Protected side vs. opposite side of the same road
- Cohorts tested before vs. after the barrier was built
- Exposure from the mother's address in ρ_b, before the road project

What we expect to see: zero, ramp, plateau

If home noise harms scores cumulatively



- Tested before the barrier ($e < 0$): $\beta_e = 0$
- Each later cohort gets one more year, at a younger age, so the step $\beta_e - \beta_{e-1}$ is the effect of one year at age $A - e$
- The step at $A + 1$ is exposure in utero
- Born after the barrier: same dose, so flat in the short run
- Later, adaptation, traffic growth or a wall on the other side may bend it: window capped at L years, slope tested

Other explanations predict other shapes



Pre-trend
sorting, anticipation



Step, no ramp
noise around the test



No plateau
trends set off by the road project

Exposure is assigned before the road project

The problem

- Families may move in, move out or stay *because of* the barrier
- Projects are announced years ahead, so current address would mix sorting into the ramp

Our rule

- Reference cell: the mother's 100 m cell on 31 December of ρ_b
- ρ_b is the year before the project was announced; until we have project dates, $\tau_b - 3$
- Robustness: $\tau_b - 1$ and $\tau_b - 5$, compared per year lived (2SLS), which dilution does not move
- Anticipation: moves out of and into the reference cell between ρ_b and τ_b, by side
- Children born after τ_b get a cell through their mother, like their older siblings
- We follow children after they move (intention to treat)

Unlike earlier work, we keep movers and follow people, not dwellings: who stays (Lindgren 2021) and who lives in a dwelling (Moretti and Wheeler 2025) both respond to the barrier.

Estimation: one stack per barrier

$$S_i = \sum_{e \neq -1} \beta_e \, \mathbf{1}\{e_i = e\} \, P_i + \gamma_{b,P_i,r(i)} + \eta_{b,e_i} + X'_{f(i)}\phi + \varepsilon_i$$

- S_i: grade-9 merit value, percentile within the test year (grading rules changed)
- Event time and dose use the *intended* test year $t^* = a_i + 16$; children who repeat or start late are scored when they take the test. Second outcome: grade 9 on time
- $\gamma_{b,P,r}$: barrier $\times$ side $\times$ distance band r (0–100, 100–250, 250–600 m, fixed in advance)
- $\eta_{b,e}$: barrier $\times$ test year, shared by both sides; absorbs traffic, local shocks and project effects common to all distances. Robustness: $\eta_{b,r,e}$
- $X_{f(i)}$: parental education, income and immigrant background at ρ_b
- No school FE, since the grade-9 school is chosen after τ_b. Robustness: *assigned* school $\times$ test year
- Each stack uses only its own barrier's cells; comparison cells that another barrier protects within the window are dropped (Cengiz et al. 2019)
- Weighted by protected children (Wing et al. 2024)
- Event study: barriers built by 2007 (full window)
- Clustered by connected corridors; wild cluster bootstrap, randomisation inference
- Main estimate: effect per year of assigned dose, if pre-trends are flat and the ramp is roughly linear
- Other estimators: Callaway and Sant'Anna (2021), Sun and Abraham (2021), Borusyak et al. (2024)

Separating noise from the road project

Barriers come with a road project and may lower air pollution (Baldauf et al. 2008; Greenhill 2026).

- Construction and traffic changes hit both sides, so the opposite side nets them out
- Air pollution partly passes over a wall, so the gap between the sides is smaller than for noise
- What remains are the side-specific parts of the project and of air pollution

Test	Rules out	If it is noise	If it is not
Rail vs. road	Air pollution	Similar ramps (trains are mostly electric)	Ramp for road barriers only
Threshold, fixed in advance: L^0 above WHO guideline (53 dB road, 54 dB rail)	Project	Ramp above, none below	Ramp also below, in quiet cells as close to the road
Intensity: predicted dB removed at each cell	Project	Larger ramp where more dB are removed	Same ramp whatever the dB cut

Same distance holds the project fixed. Air pollution moves with distance and traffic like noise; only rail vs. road tests it.

Backup

Backup: overview

Slide	What it covers	Backs up
Threats and answers →	Each threat, the design answer and the evidence we show	All
Distance and comparison groups →	The three comparison groups; effect by 50 m bin; balance before the barrier	Design
Mobility and per year lived →	Moving as a mechanism; 2SLS on years actually lived	Assignment
Dose and age profile →	Parametric effect per year of dose; kinks by age; duration against intensity	Prediction, estimation
Other estimands and checks →	Siblings, residents at the test, heterogeneity, placebo checks	Threats
Literature →	What we take from each study and what we do differently	Contribution
References →		

Backup: threats and answers

Threat	Design answer	Evidence we show
Families sort or anticipate	Reference cell at ρ_b, via the mother; ITT	Mobility event study; births by side; predicted-score placebo
Road project comes with the barrier	Opposite side of the same road; barrier $\times$ year FE	Threshold, intensity, plateau; barrier $\times$ band $\times$ year FE
Air pollution moves with noise	Rail against road	NO_2 by side, if a grid exists
Staggered timing, bad comparisons	Stacking, own cells only, corrective weights	Callaway–Sant'Anna, Sun–Abraham, Borusyak et al.
Cells misclassified at the zone edge	Drop edge cells (100 m grid)	Estimates with them included
Comparison cells insulated	— (bias toward zero)	Insulation records, if they exist
Grading changes	Percentile within test year	Raw merit value, national tests
Sides feed different schools; grading by school	Assigned school $\times$ year FE (not actual school: post-treatment)	School overlap of the two sides; actual school $\times$ year FE as decomposition

◂ Overview

Backup: distance profile and comparison groups

Three comparison groups

1. Opposite side, same distance (main)
2. 600–1,500 m behind the barrier
3. Not-yet-treated corridors (robustness)

- We do not rely on never-treated places alone: places that qualified but were never shielded may have low priority for reasons that also shape their trends
- The opposite side may have no barrier because it has few dwellings or was insulated; we check this

Distance profile

- Effect in 50 m bins up to 1,500 m from the barrier, each against the opposite side at the same distance
- Shows how far protection reaches and whether the 600–1,500 m band is unaffected; if it is not, the band moves out
- Balance: before the barrier, scores should vary with distance in the same way on both sides (Berkhout et al. 2025)

◂ Overview

Backup: mobility and the effect per year lived

$$M_{ft} = \sum_{k \neq -1} \mu_k \, \mathbf{1}\{t - \tau_b = k\} \, P_f + \gamma_{b,P_f,r(f)} + \eta_{b,t} + X_f' \phi + u_{ft}$$

- Mobility: hazard of leaving the reference cell. Moves between ρ_b and τ_b would show anticipation
- Per year lived: 2SLS on years actually lived behind a barrier, instrumented with the assigned dose. This is an average causal response among compliers (Angrist and Imbens 1995)
- It is a *LATE* if the barrier does not change moving or schools, otherwise a *rescaled ITT*. Bounds allow a direct effect of assignment (Conley et al. 2012)
- Compliers described by household characteristics at ρ_b (Abadie 2003)

◂ Overview

Backup: effect per year of dose and age profile

$$S_i = \beta\, D_i P_i + \beta^{\text{utero}}\, \mathbf{1}\{e_i > A\}\, P_i + \beta^{\text{pre}}\, \mathbf{1}\{e_i < 0\}\, e_i\, P_i + \gamma_{b,P_i,r(i)} + \eta_{b,e_i} + X'_{f(i)}\phi + \varepsilon_i$$

$$D_i = \min\{\max(0,\ e_i),\ A\}$$

- β is the effect per year of assigned dose, β^{utero} the added effect of exposure in pregnancy, β^{pre} a pre-trend test
- Age profile: the slope may differ by the age of the added year, with kinks at 3, 7 and 13
- Why not regress on D_i alone? Cohorts tested before would pool with comparison cells, and we would lose the placebo. A continuous dose with staggered timing also needs stronger parallel trends (Callaway et al. 2024)
- Duration vs. intensity: years of exposure, holding dB-years $D_i \Delta L$ fixed (Hollingsworth et al. 2025); ΔL permuted within corridors

◂ Overview

Backup: other estimands and checks

Other estimands

- Siblings: household FE with birth order $\times$ sex FE (Black et al. 2005). The sample leans toward large families that stay, so robustness only
- Residents at the test: the dwelling-based estimand. Its gap to the ITT measures sorting; test for compositional change (Sant'Anna and Xu 2025)

Heterogeneity, fixed in advance

- Sex; parental education, income, immigrant background; housing type
- Wind direction: heterogeneity by side only, since it carries pollution too

Checks

- Predicted score from household characteristics at ρ_b as a placebo outcome
- Births after ρ_b: number and mothers' characteristics by side
- τ_b shifted back five years
- Off-age testing by P_i after τ_b; children tested on time only
- Dated against undated barriers: location, length, road type, neighbourhood
- Birth outcomes for cohorts born after τ_b (access to be confirmed)

◂ Overview

Backup: how this relates to the literature

Study	We take	We do differently
Lindgren (2021): Swedish noise mitigation and health	The setting	Keep movers; same-road controls, not never-treated
Moretti and Wheeler (2025): barriers and house prices	The 600–1,500 m band	Follow people, not dwellings; no unbuilt barriers in Sweden
Greenhill (2026): noise and infant health	Noise and air pollution move together	Rail as a near noise-only treatment
Grönqvist et al. (2020): lead, Swedish cohorts	Cohort × exposure, threshold fixed in advance, percentile outcome	Place-specific timing; dose from years lived
Hollingsworth et al. (2025); Berkhout et al. (2025)	Duration against intensity; age profile; distance cut-off from the data	Noise, not lead

We add a cohort design for noise and school achievement: exposure assigned before the barrier, compared within each corridor.

◂ Overview

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